~/majibill/docs/08-product-ux-spec.md

$ cat 08-product-ux-spec.md

MAJIBILL PLATFORM · PRODUCT

Product, UX & Module Specification

Personas, module scope by phase, screen inventory, key journeys, wireframes, design principles and reporting for the MajiBill web and field apps.

```
// dense tables, keyboard-first, fast on slow networks
// the reader works offline; the clerk works in bulk
// every number links to the records behind it
```

DOC
MB-08

VERSION
1.0 draft

DATE
08 Oct 2026

STATUS
For review

AUDIENCE
Product, design, frontend, mobile, client stakeholders

CLASS
Confidential

Prepared by 404HubSpot · Nakuru, Kenya · https://404-hub-spot.vercel.app
Working codename: MajiBill (Swahili: maji = water). Figures marked 'assumption' must be validated before commitment.

// Contents

1. Product intent 3
2. Module scope by phase 3
3. Screen inventory 4
4. Key journeys 4
5. Wireframes 5
6. UX and design principles 6
7. Notifications and templates 6
8. Reports 6

// 1. Product intent

Give water utilities a billing and revenue-collection system that is simple to adopt, correct by construction, and fit for Kenyan realities: M-Pesa first, offline field work, SMS everywhere, low-end devices, and auditors who ask hard questions.

1.1 Personas

Persona	Job to be done	Primary surface	Pain today
Billing clerk	Register customers, review readings, fix errors, issue bills	Staff web app	Spreadsheets, double entry, no audit
Meter reader	Read every meter on a route and prove it	Reader app (offline)	Paper books, lost sheets, disputed reads
Cashier / accountant	Post cash and reconcile M-Pesa and bank	Staff web app	Unmatched payments, manual reconciliation
Utility manager	See collections, arrears, losses, performance	Dashboards, reports	Late and inconsistent numbers
Auditor / regulator liaison	Trace and verify	Audit viewer, exports	No trail
Customer	Know the bill, pay easily, get a receipt, report issues	SMS, portal, M-Pesa	Surprise bills, no proof of payment
Platform admin (404HubSpot)	Onboard and run tenants	Platform console	—
Partner admin (ZUCAR)	Sell, onboard, support its customers	Partner portal	—

// 2. Module scope by phase

The ZUCAR deck defines the **Standard** package. Several items it lists as future enhancements are essential at county scale, so the phasing below pulls them forward.

Module	MVP (pilot)	v1.1	v2
Customer & account management	Register, edit, status, search, import	Bulk ops, documents	Customer self-registration
Meters & routes	Meter register, install/replace, routes, zones	Meter inventory/stock	Smart meter (AMR) ingest
Readings	Web + offline mobile capture, validation, estimates, photo evidence	Supervisor review queue, GPS rules	Photo OCR assist
Tariffs & charges	Block tariffs, fixed, meter rent, sewer, versions	Penalty, rebate rules, special tariffs	Rule designer UI
Billing	Cycles, bill run, approval, bills, adjustments	Reissue, batch print, bill templates	Prepaid / smart billing
Payments	M-Pesa C2B auto-post, cash, suspense, receipts	STK push from portal, bank CSV import	Cards, bank API, USSD
Statements & notices	Statements, SMS bills and receipts, reminders	Email, WhatsApp-style channels	Multilingual templates
Reports	Arrears, collections, consumption, revenue, WASREB-aligned KPIs	Scheduled reports, exports	Analytics: NRW, anomalies, forecasts
Admin & security	Users, roles, audit viewer, settings, branding	SSO, API keys, webhooks	Marketplace of integrations
Platform console	Tenants, plans, cells, support sessions	Partner portal, usage billing	Self-serve onboarding

// 3. Screen inventory

Area	Screens
Auth	Login, MFA, forgot password, session expired
Dashboard	Collections today/month, billed vs collected, arrears ageing, flagged readings, suspense payments, pending approvals
Customers	List + filters, customer profile (accounts, meters, bills, payments, notes, audit), create/edit, bulk import wizard
Accounts	Account detail with balance explainer, statements, status actions (suspend/disconnect/reconnect/close)
Meters & routes	Meter list, meter detail (history, installs), replace-meter flow, route builder, assignment board
Readings	Capture (web), review queue, anomalies, corrections, bulk estimate, route progress map
Billing	Cycle list, cycle detail (stages), bill run monitor, review/approve, bill viewer, adjustments queue
Payments	Payments list, record cash, M-Pesa feed, suspense workbench, reconciliation, receipts, reversals
Reports	Report catalogue, parameter form, results table/chart, export, scheduled reports
Admin	Users, roles & permissions, tariffs editor, charges, templates, branding, integrations, audit viewer
Customer portal	Login by phone OTP, balance, bills, pay now (STK), receipts, usage chart, report a problem
Platform console	Tenants, partners, plans, cells, fleet health, support sessions, usage and invoicing

// 4. Key journeys

Reading to bill (clerk’s view)

1. Open cycle → assign routes to readers → readers sync and capture offline.
2. Dashboard shows route progress; flagged readings land in the review queue.
3. Clerk clears flags (accept, correct with evidence, estimate). Cycle is frozen.
4. Run bill calculation; monitor progress; review exceptions; supervisor approves.
5. Publish: bills generated, SMS queued, portal updated, ledger posted.

Customer pays by M-Pesa

1. Customer receives SMS: amount, due date, Paybill and account number.
2. Pays on M-Pesa; confirmation callback posts payment within seconds.
3. Customer receives receipt SMS with new balance; clerk sees the payment instantly; unmatched ones appear in the suspense workbench.

// 5. Wireframes

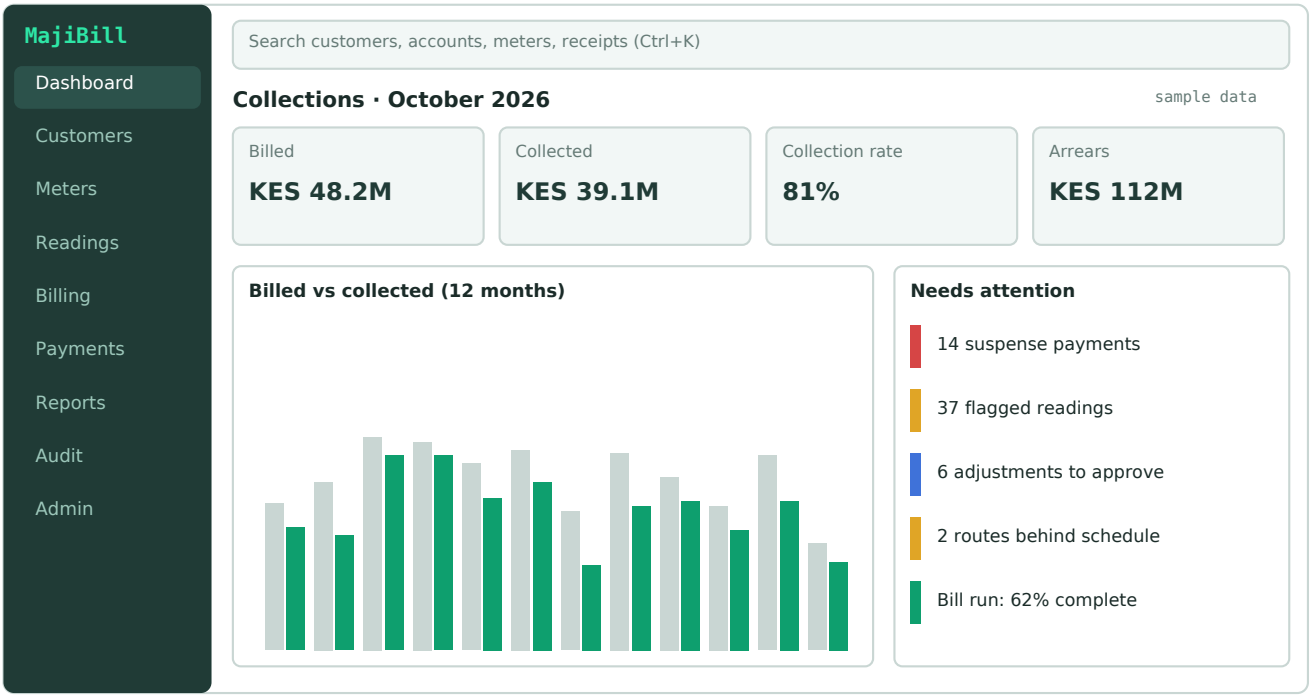


Fig 5.1 Staff dashboard wireframe (illustrative data). Every tile drills into the records behind it.

Reader app: route, capture, sync

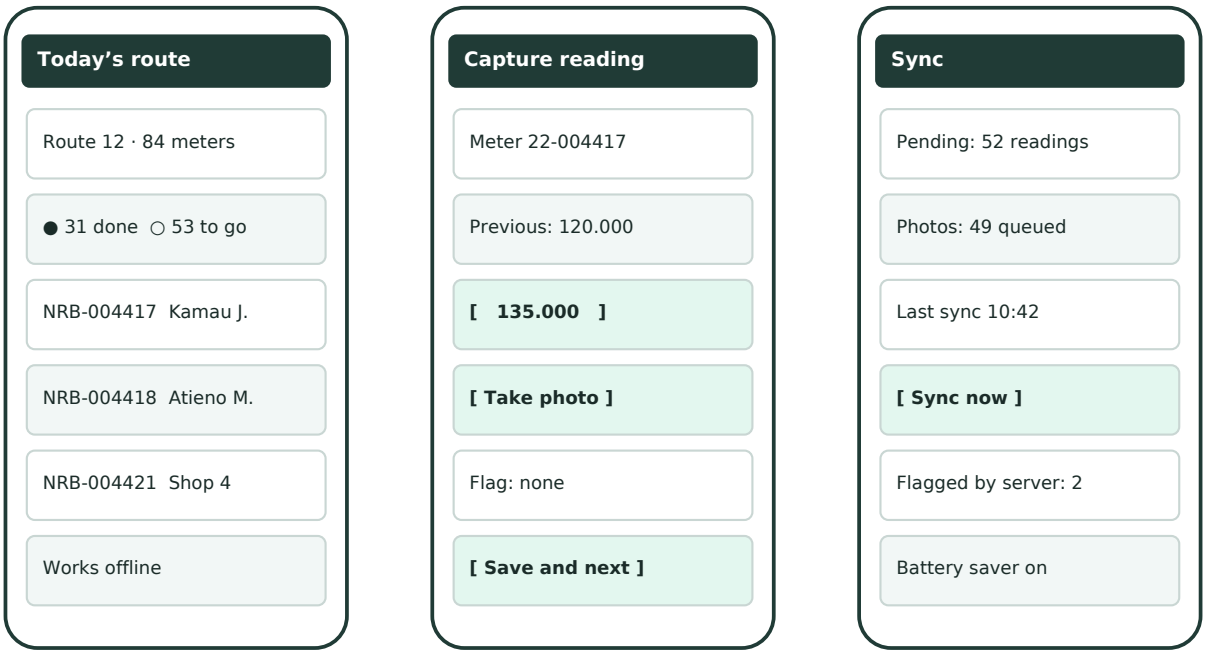


Fig 5.2 Meter-reader flow. Big touch targets, one primary action per screen, offline by default.

// 6. UX and design principles

Principle	What it means
Dense, fast, keyboard-first	CRM-style tables with saved filters, bulk actions, inline edit, global search, shortcuts. Clerks process hundreds of rows a day.
Explain every number	Balances and totals link to bills, payments and audit events behind them.
Exceptions first	Home screens show what needs attention (flags, suspense, approvals), not vanity charts.
Built for low bandwidth	Small bundles, server-side pagination, optimistic UI, offline reader, SMS as the universal channel.
Safe by design	Confirmations for irreversible actions, maker–checker prompts, clear reasons required, undo via reversal.
Local fit	KES formatting, E.164 phone input with Kenyan defaults, English and Kiswahili, East Africa Time, SMS length limits.
Accessible	WCAG 2.2 AA targets, high contrast, large touch targets in the field app, no colour-only status.
White-label	Per-tenant logo, colours, bill templates, sender ID, support contacts.

6.1 Design system

- Web: Next.js 16, React 19, Tailwind CSS 4 and shadcn/ui, built from the team's `sales-crm` starter and its conventions file as the base for tables, forms, filters and layout; CRM pipeline patterns inform the bill-run and approval views. Final component choices are confirmed in a design review.
- Tokens: dark green primary, mint accent, neutral greys (aligned with the 404HubSpot brand); light and dark themes; status colours with icons.
- Components built once in `packages/ui`: DataTable, FilterBar, MoneyInput, PhoneInput, StatusBadge, Timeline, DiffView, ApprovalBar, EmptyState.
- Mobile: Flutter with Material 3, high-contrast theme, offline indicators, large inputs.
- Every screen has loading, empty, error, and permission-denied states specified before build.

// 7. Notifications and templates

Event	Channel	Example (draft)
Bill issued	SMS, portal, email	Dear {name}, water bill {bill_no}: KES {amount}, due {date}. Pay via M-Pesa Paybill {paybill}, Acc {account_no}.
Payment received	SMS, portal	Received KES {amount} on {date}. Receipt {receipt_no}. Balance KES {balance}. Thank you.
Reminder	SMS	Reminder: KES {amount} due {date}. Pay to avoid disconnection.
Disconnection notice	SMS + letter	Final notice per utility policy; includes dispute contact.
Reading received	Portal	Your meter reading {value} m³ was recorded on {date}.

Templates are tenant-configurable, versioned and audited; SMS segments counted and costed before bulk send.

// 8. Reports

Reports align with the sector indicators the regulator (WASREB) tracks, so tenants can produce regulatory returns directly: water coverage inputs, non-revenue water, metering ratio, collection efficiency, revenue vs operating cost coverage, staff productivity and service hours where data exists.

Report	Audience
Arrears ageing (30/60/90+), by zone and class	Finance, manager
Collections: daily, by channel, by cashier, reconciliation status	Finance
Billed vs collected; collection efficiency	Manager, regulator return

Report	Audience
Consumption analysis, high/low/zero consumers	Operations
Non-revenue water (needs bulk/production meters)	Operations, regulator return
Metering ratio and reading coverage by route	Operations
Suspense and exceptions; write-offs; adjustments	Finance, auditor
Audit and access reports	Auditor, platform security